Supplementary derivation (standalone build)

This section derives every relation the screen evaluates. Each part opens with the physical argument; the algebra follows it. The notation is that of the symbol register of the theory package (listed in the parameter tables), and every displayed relation carries its SI units in brackets. Two regimes are kept apart throughout: the water-processed, alkaline electrode slurry at room temperature (R-slurry) and the non-aqueous, fluorine-free cell with the Al collector at the cathode's potential (R-cell). A relation is used only in the regime whose reference states it assumes. The links C1–C5 name what the proposed mechanism must achieve: anchor the template (C1), bind Ca^{2+} (C2), nucleate the mineral (C3), form a continuous film (C4) and keep it stable (C5). Meeting the conditions below makes the mechanism thermodynamically possible under the stated framing; it does not show that a barrier works, and no rate, capacity or device statement follows from them.

0.1 Adhesion of the template in water (C1)

Physics. In the slurry the template never meets a dry oxide. The hydroxylated oxide and the polymer are both covered by water, and a contact between them forms only by displacing that water. The competing states are therefore the contact and the two water-wetted surfaces, not the contact and two bare surfaces.

The Dupré work to separate a contact between the oxide (o) and the polymer (p) in water (w) creates an oxide–water and a polymer–water interface and destroys the oxide–polymer one:

$$\begin{aligned} W_{\mathrm{adh,wet}} &= \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - \gamma_{\mathrm{op}} = W_{\mathrm{adh,dry}} + 2\gamma_{\mathrm{w}} - W_{\mathrm{ow}} - W_{\mathrm{pw}}, \\ W_{\mathrm{adh,dry}} &= \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}], \end{aligned} \tag{S1}$$

where $W_{\mathrm{ow}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{w}} - \gamma_{\mathrm{ow}}$ and $W_{\mathrm{pw}} = \gamma_{\mathrm{p}} + \gamma_{\mathrm{w}} - \gamma_{\mathrm{pw}}$ are the works with which water binds each surface. A large dry adhesion is offset by those works, so $W_{\mathrm{adh,wet}}$ can be small or negative for a hydrophilic oxide and polymer. The bare oxide has $W_{\mathrm{adh,wet}} = 0$ and fixes the zero. The route that decides C1 is the potential of mean force of a template fragment in explicit water, $W_{\mathrm{adh,wet}} = \Delta F_{\mathrm{PMF}}/A_{\mathrm{c}}$, which contains the displacement penalty and the entropy of the released water; it needs a PMF that plateaus at large separation and one contact-area definition for all fragments.

C1 inputs in P0. No PMF exists for P0, so the registered C1 verdict is a gap and counts as inconclusive. Two estimates are reported beside it. (i) A flagged analog: works of adhesion in water measured for catechol-bearing films on other oxides. A pull-off measurement records a work only for contacts that adhere, so the analog has positive support and cannot represent displacement by water; its verdict is conditional on the analog. (ii) The Dupré form with ignorance-bounded energies, $\gamma_{\mathrm{ow}} \in [0, \gamma_{\mathrm{o}} + \gamma_{\mathrm{w}}]$ and $\gamma_{\mathrm{op}} \in [0, \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}}]$ (Section 0.9), and γ_{pw} the sourced template–water energy truncated at zero. It places $W_{\mathrm{adh,wet}}$ in $[\gamma_{\mathrm{pw}} - \gamma_{\mathrm{o}} - \gamma_{\mathrm{p}},\ \gamma_{\mathrm{pw}} + \gamma_{\mathrm{o}} + \gamma_{\mathrm{w}}]$. Since $\gamma_{\mathrm{pw}} \le \gamma_{\mathrm{p}} + \gamma_{\mathrm{w}}$, the lower end is at most $\gamma_{\mathrm{w}} - \gamma_{\mathrm{o}}$, negative whenever the dry oxide's surface energy exceeds water's: thermodynamic bounds alone admit both signs of the wet adhesion, which is why a PMF is needed. Masses under these bounds are properties of the prior, never a finding; the flip value $\gamma^{*}_{\mathrm{op}} = \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - m_W$ is reported instead.

0.2 The driving force: uniform chemical potential

Physics. A nucleus is built from ions taken out of the solution, and it gains, per formula unit, the chemical potential the ions carry minus that of the solid. The template sits in an open interfacial region that exchanges ions with the slurry. At equilibrium nothing is gained by moving an ion between bulk and layer, so every ion carries the same electrochemical potential everywhere. The template can concentrate ions; it cannot make them carry more free energy.

One definition. With K_{sp} the solubility product of the registered framing (calcite and amorphous $CaCO_3$ are both registered, each run as a whole chain with its own K_{sp}, V_{m} and γ_{nl}),

$$\Omega = \frac{a_{\mathrm{Ca}^{2+}}\, a_{\mathrm{CO}_3^{2-}}}{K_{\mathrm{sp}}} \quad [1], \qquad \Delta\mu = k_{\mathrm{B}}T \ln \Omega \quad [\mathrm{J}], \tag{S2}$$

per $CaCO_3$ formula unit, with molal activities. Nucleation requires $\Omega > 1$. For a hydrated framing the ion activity product carries a_{w}^{n}; keeping $a_{\mathrm{Ca}^{2+}} a_{\mathrm{CO}_3^{2-}}$ overstates Ω by a_{w}^{-n} at a drying state ($a_{\mathrm{w}} < 1$), so drying states are optimistic, a_{w} is reported there, and the hydrated ratio is a labelled sensitivity.

Equilibrium of the open region. The electrochemical potential of ion i at $\mathbf{r}$ is

$$\tilde{\mu}_i(\mathbf{r}) = \mu_i^{\circ} + w_i^{\mathrm{sp}}(\mathbf{r}) + k_{\mathrm{B}}T \ln \frac{\gamma_i(\mathbf{r}) m_i(\mathbf{r})}{m^{\circ}} + z_i e \psi(\mathbf{r}) \quad [\mathrm{J}], \tag{S3}$$

with w_i^{sp} the specific interaction with the template, γ_i the non-specific activity coefficient and ψ the mean potential (both w^{sp} and ψ vanish in bulk). Equilibrium with the slurry reservoir is $\tilde{\mu}_i(\mathbf{r}) = \tilde{\mu}_{i,\mathrm{bulk}}$ for every ion and every $\mathbf{r}$. Solving for m_i gives the local enrichment $m_i/m_{i,\mathrm{bulk}} = (\gamma_{i,\mathrm{bulk}}/\gamma_i) \exp[-(w_i^{\mathrm{sp}} + z_i e\psi)/k_{\mathrm{B}}T]$: the Boltzmann enrichment is a consequence of uniform potentials, not an exception to them. The driving force for adding a formula unit at $\mathbf{r}$ is then

$$\Delta\mu(\mathbf{r}) = \tilde{\mu}_{\mathrm{Ca}}(\mathbf{r}) + \tilde{\mu}_{\mathrm{CO_3}}(\mathbf{r}) - \mu_{\mathrm{CaCO_3(s)}} = k_{\mathrm{B}}T \ln \Omega_{\mathrm{bulk}} \quad [\mathrm{J}], \tag{S4}$$

the same at a binding site, in the diffuse layer and in bulk.

Specific binding. At a site, binding lowers the standard part of $\tilde{\mu}_{\mathrm{Ca}}$ by $-\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$ and raises the concentration term by $k_{\mathrm{B}}T \ln \mathcal{E} = -\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$. The two cancel:

$$\Delta\mu_{\mathrm{site}} - \Delta\mu_{\mathrm{bulk}} = k_{\mathrm{B}}T \ln \mathcal{E} + \Delta G_{\mathrm{seq}}/N_{\mathrm{A}} = 0 \quad [\mathrm{J}]. \tag{S5}$$

A mean potential. At an anionic template Ca^{2+} is enriched by $\mathrm{e}^{-2e\psi/k_{\mathrm{B}}T}$ and CO_3^{2-} depleted by $\mathrm{e}^{+2e\psi/k_{\mathrm{B}}T}$. The ion product is unchanged, and in Eq. (S4) the potential enters as $(z_{\mathrm{Ca}} + z_{\mathrm{CO_3}})e\psi = 0$ because the solid is neutral.

Result. The nucleus at the template is driven by $k_{\mathrm{B}}T \ln \Omega_{\mathrm{bulk}}$ at the regime activities. The template changes the barrier only through interfacial energies. A local driving force above the bulk one needs a non-equilibrium state with a time scale (drying faster than exchange, a layer loaded at another activity, a local pH excursion); P0 makes no statement about such states.

0.3 Carbonate speciation and the activity model

Physics. Ω needs the activities of the free ions. In an alkaline, carbonate-bearing slurry part of the calcium and carbon is bound in ion pairs and part of the carbon is bicarbonate or dissolved CO_2, so the totals overstate the free ions. The registered state fixes pH, the dissolved inorganic carbon C_{T}, the total dissolved calcium $\mathrm{Ca_T}$ and the ionic strength I, in a closed solution that has not yet precipitated.

With $a_{\mathrm{H}} = 10^{-\mathrm{pH}}$ and $y \equiv a_{\mathrm{CO}_3^{2-}}$, the thermodynamic constants (at $I = 0$) tie every species to y and $a_{\mathrm{Ca}^{2+}}$: $a_{\mathrm{HCO_3}} = 10^{\mathrm{p}K_2 - \mathrm{pH}} y$, $a_{\mathrm{CO_2}} = 10^{\mathrm{p}K_1 + \mathrm{p}K_2 - 2\mathrm{pH}} y$, $a_{\mathrm{CaCO_3^0}} = K_{\mathrm{CaCO_3^0}} a_{\mathrm{Ca}^{2+}} y$, $a_{\mathrm{CaHCO_3^+}} = K_{\mathrm{CaHCO_3^+}} a_{\mathrm{Ca}^{2+}} a_{\mathrm{HCO_3}}$ and $a_{\mathrm{CaOH^+}} = {}^{*}K_{\mathrm{CaOH^+}} a_{\mathrm{Ca}^{2+}} a_{\mathrm{w}}/a_{\mathrm{H}}$. Dividing each activity by its coefficient ($\gamma_{\pm 1}$ or $\gamma_{\pm 2}$ by charge; one for neutral species) and summing gives the carbon and calcium balances in units of the standard molality:

$$m_{C_{\mathrm{T}}} = A_{\mathrm{c}} y + P_{\mathrm{c}} a_{\mathrm{Ca}^{2+}} y, \qquad m_{\mathrm{Ca_T}} = D_{\mathrm{c}} a_{\mathrm{Ca}^{2+}} + P_{\mathrm{c}} a_{\mathrm{Ca}^{2+}} y \quad [1], \tag{S6}$$

$$
\begin{aligned}
A_{\mathrm{c}} &= 10^{\mathrm{p}K_1+\mathrm{p}K_2-2\mathrm{pH}} + \frac{10^{\mathrm{p}K_2-\mathrm{pH}}}{\gamma_{\pm1}} + \frac{1}{\gamma_{\pm2}},\\
P_{\mathrm{c}} &= K_{\mathrm{CaCO_3^0}} + \frac{K_{\mathrm{CaHCO_3^+}}\,10^{\mathrm{p}K_2-\mathrm{pH}}}{\gamma_{\pm1}},\\
D_{\mathrm{c}} &= \frac{1}{\gamma_{\pm2}} + \frac{{}^{*}K_{\mathrm{CaOH^+}}\,a_{\mathrm{w}}10^{\mathrm{pH}}}{\gamma_{\pm1}} \quad [1],
\end{aligned}
\tag{S7}
$$

with $m_{C_{\mathrm{T}}} = C_{\mathrm{T}}/(\rho_{\mathrm{w}}m^{\circ})$ and $m_{\mathrm{Ca_T}} = \mathrm{Ca_T}/(\rho_{\mathrm{w}}m^{\circ})$ in dilute solution. Eliminating $a_{\mathrm{Ca^{2+}}} = m_{\mathrm{Ca_T}}/(D_{\mathrm{c}} + P_{\mathrm{c}}y)$ leaves $A_{\mathrm{c}}P_{\mathrm{c}}y^2 + By - m_{C_{\mathrm{T}}}D_{\mathrm{c}} = 0$ with $B = A_{\mathrm{c}}D_{\mathrm{c}} + P_{\mathrm{c}}(m_{\mathrm{Ca_T}} - m_{C_{\mathrm{T}}})$. The product of the roots is negative, so exactly one root is positive; in the form that stays exact as $P_{\mathrm{c}} \to 0$,

$$
a_{\mathrm{CO_3^{2-}}} = \frac{2m_{C_{\mathrm{T}}}D_{\mathrm{c}}}{B + \sqrt{B^2 + 4A_{\mathrm{c}}P_{\mathrm{c}}m_{C_{\mathrm{T}}}D_{\mathrm{c}}}}, \qquad a_{\mathrm{Ca^{2+}}} = \frac{m_{\mathrm{Ca_T}}}{D_{\mathrm{c}} + P_{\mathrm{c}}a_{\mathrm{CO_3^{2-}}}} \quad [1]. \tag{S8}
$$

With no ion pairs or no calcium, $a_{\mathrm{CO_3^{2-}}} = m_{C_{\mathrm{T}}}/A_{\mathrm{c}}$, the carbonate system alone; pairs can only lower it. The free $\mathrm{Ca^{2+}}$ concentration $c_{\mathrm{bulk}} = a_{\mathrm{Ca^{2+}}}\rho_{\mathrm{w}}m^{\circ}/\gamma_{\pm2}$ is the c_{bulk} of the binding relations below: the free ion, not the total.

Activity model. Davies: $\log_{10}\gamma_{\pm z} = -A_{\mathrm{DH}}z^2\big[s/(1+s) - b_{\mathrm{D}}s^2\big]$ with $s = \sqrt{I/m^{\circ}}$, so $\gamma_{\pm2} = \gamma_{\pm1}^4$ and every coefficient tends to one as $I \to 0$. The slope $\mathrm{d}\log_{10}\gamma/\mathrm{d}s$ changes sign at the positive root s_* of $2b_{\mathrm{D}}s(1+s)^2 = 1$; beyond $I_* = m^{\circ}s_*^2$ the ionic-strength dependence is carried by the empirical linear term alone. The registered validity bound therefore satisfies $I_{\mathrm{val}} \le I_*$, and a draw with $I > I_{\mathrm{val}}$ is refused and counted, never extrapolated. Neutral species take coefficient one, and I is the registered input rather than recomputed from the speciation; both are stated limitations.

0.4 $\mathrm{Ca^{2+}}$ binding at the template (C2, a descriptor)

Physics. The template can raise the local $\mathrm{Ca^{2+}}$ concentration only if it holds $\mathrm{Ca^{2+}}$ more strongly than water does. The free energy must start from hydrated $\mathrm{Ca^{2+}}$ at the regime activity, pay for preparing the site at the regime pH, and end in a well-defined local state.

For an acid site L–H that binds only when deprotonated, the cycle (deprotonate at the regime pH, then transfer $\mathrm{Ca^{2+}}$ from bulk to the site) gives

$$
\begin{aligned}
\Delta G_{\mathrm{seq}} &= \Delta G_{\mathrm{site}} + RT\ln\left(1 + 10^{\,\mathrm{p}K_{\mathrm{a}}-\mathrm{pH}}\right),\\
\Delta G_{\mathrm{site}} &\equiv -RT\ln\left[\frac{1}{V_{\mathrm{site}}}\int_{\mathrm{site}} \mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}\,\mathrm{d}^3r\right] \quad [\mathrm{J\,mol^{-1}}],
\end{aligned}
\tag{S9}
$$

with w the PMF of one ion relative to bulk and V_{site} the binding-layer volume per site, fixed once. A binding constant to a standard concentration c° compares with ΔG_{site} only after $\Delta G^{\circ}_{\mathrm{bind}} = \Delta G_{\mathrm{site}} - RT\ln(N_{\mathrm{A}}c^{\circ}V_{\mathrm{site}})$. In the dilute limit with non-interacting sites the enrichment is the concentration ratio $\mathcal{E} = c_{\mathrm{local}}/c_{\mathrm{bulk}} = \exp(-\Delta G_{\mathrm{seq}}/RT)$; by Section 0.2 it never enters Ω. For Γ_{s} sites per area in a layer δ_{s} thick (Langmuir),

$$
\begin{aligned}
x &= \frac{\mathcal{E}c_{\mathrm{bulk}}}{c_{\mathrm{max}}}, \quad c_{\mathrm{max}} = \frac{\Gamma_{\mathrm{s}}}{\delta_{\mathrm{s}}}, \quad \Theta = \frac{x}{1+x},\\
c_{\mathrm{local}} &= c_{\mathrm{max}}\Theta \le \min(\mathcal{E}c_{\mathrm{bulk}}, c_{\mathrm{max}}),\\
\Gamma_{\mathrm{Ca}} &= \Gamma_{\mathrm{s}}\Theta - \delta_{\mathrm{s}}c_{\mathrm{bulk}} \quad [1;\ \mathrm{mol\,m^{-2}}],
\end{aligned}
\tag{S10}
$$

where Γ_{Ca} is the Gibbs excess: the bound amount per area less what the layer volume holds at the bulk concentration. A strongly binding but sparse template is limited by its site density. These quantities, and the adsorption term of the next part, are what C2 reports. C2 does not gate: its only route to nucleation is the template–liquid energy, which C3 already judges. The mechanism's premise, $\Delta G_{\mathrm{seq}} < 0$, is reported as a finding (it holds, fails or is undecided under the shared decision rule).

Geometric bounds on Γ_s and δ_s. Each site holds one ion, and two bound ions cannot be closer than their contact distance $2r_{Ca}$. The densest planar packing of equal discs gives each the area $2\sqrt{3}\,r^2$, so the footprint of a site obeys $a_{site} \geq 2\sqrt{3}\,r_{Ca}^2$ (one interfacial layer of sites). A volume of template of mass density ρ_{PDA} holds at most ρ_{PDA}/M_u moles of coordinating groups per volume, with M_u the least template mass per mole of groups. Hence

$$\begin{aligned} 0 < \Gamma_s &\leq \min\Big(\frac{1}{2\sqrt{3}\,N_A r_{Ca}^2},\ \frac{\rho_{PDA} d_{layer}}{M_u}\Big), \\ \max\big(2r_{Ca},\ \Gamma_s V_m\big) \leq \delta_s &\leq d_{layer} \quad [\mathrm{mol\,m^{-2}};\ \mathrm{m}]. \end{aligned} \tag{S11}$$

The layer must contain the ion ($2r_{Ca}$), cannot hold Ca^{2+} more densely than a solid Ca phase ($\Gamma_s/\delta_s \leq 1/V_m$) and lies inside the template layer. r_{Ca} is the ionic radius at the lowest registered coordination number, the smallest admissible, so the bound holds at any higher coordination. The Ca–O distance of the hydrated ion exceeds r_{Ca} and would tighten both bounds beyond what geometry allows, so it is used only as a labelled sensitivity; while r_{Ca} is unsourced the $2r_{Ca}$ term is dropped, which keeps the bound valid.

0.5 Where the bound Ca^{2+} goes: the Gibbs–Langmuir term

Physics. Since the driving force is the bulk one, the only places left for the template's binding are the interfacial energies. Bound ions are an interfacial excess, and the Gibbs adsorption equation says how such an excess changes the energy of its interface.

Template–liquid interface. At fixed T, $\mathrm{d}\gamma_{sl} = -\sum_i \Gamma_i \mathrm{d}\mu_i$. Taking the adsorbing component as Ca^{2+} with its compensating charge (a neutral solute) and raising the Ca^{2+} activity from zero (the Ca-free template, $\gamma_{sl}^{(0)}$) to the regime value, with the excess of Eq. (S10) and $\mathrm{d}\mu_{Ca} = RT\,\mathrm{d}\ln c$,

$$\begin{aligned} \Delta\gamma_{ads} \equiv \gamma_{sl} - \gamma_{sl}^{(0)} &= -\int_0^{c_{bulk}} \Gamma_{Ca}(c)\, RT \frac{\mathrm{d}c}{c} \\ &= -RT\big[\Gamma_s \ln(1+x) - \delta_s c_{bulk}\big] \quad [\mathrm{J\,m^{-2}}], \end{aligned} \tag{S12}$$

because $x(c)$ is linear in c. Its properties: $\partial\Delta\gamma_{ads}/\partial\Delta G_{seq} = \Gamma_s\Theta > 0$, so deeper binding lowers γ_{sl}; $\partial\Delta\gamma_{ads}/\partial\mu_{Ca} = -\Gamma_{Ca}$; a repelling template has $\Delta\gamma_{ads} \to RT\delta_s c_{bulk} > 0$; at strong binding $\Delta\gamma_{ads} \to \Gamma_s\Delta G_{seq} + RT\Gamma_s \ln(c_{max}/c_{bulk}) + RT\delta_s c_{bulk}$, whose last term survives but is small when $\delta_s c_{bulk} \ll \Gamma_s$. If γ_{sl} is computed at the regime activities, $\Delta\gamma_{ads}$ is already inside it and is not added.

The neutral-solute approximation and the double-layer limitation. The cancellation of the mean potential in Section 0.2 concerns the ion product, not γ_{sl}. Binding Ca^{2+} to anionic sites changes the surface charge density σ and with it the diffuse-layer free energy, which is part of γ_{sl}; for an electrified interface the Gibbs equation carries a Lippmann term $-\sigma\mathrm{d}\psi_0$. Equation (S12) omits it. In the low-potential Gouy–Chapman limit the diffuse layer holds $\sigma^2/(2\varepsilon_r\varepsilon_0\kappa_D)$ per area, with $\kappa_D^2 \propto I$; at a fixed change of charge the omitted term falls as $I^{-1/2}$ and vanishes when binding leaves the charge unchanged. It is an unquantified correction of unassigned sign, largest at low ionic strength for a template whose charge binding changes strongly; an in-situ γ_{sl} with explicit ions contains it.

Template–nucleus interface. γ_{sn} is a grand-canonical excess at the regime chemical potentials as well. If the contact exchanges Ca^{2+} with the reservoir, any Ca^{2+} excess $\Gamma_{Ca,sn}$ at the contact (an ion bridge between a carboxylate and the crystal, say) is inside it, and

$$\frac{\partial\gamma_{sn}}{\partial\mu_{Ca}} = -\Gamma_{Ca,sn}, \qquad \frac{\partial(\gamma_{sl}-\gamma_{sn})}{\partial\mu_{Ca}} = -(\Gamma_{Ca,sl} - \Gamma_{Ca,sn}) \quad [\mathrm{mol\,m^{-2}}], \tag{S13}$$

with $\Gamma_{\mathrm{Ca,sl}} \equiv \Gamma_{\mathrm{Ca}}$. A contact sealed from the solution holds a fixed amount instead, and the relation does not apply. The screen takes γ_{sn} from contact data at the regime state and never adds ΔG_{seq} or $\Delta\gamma_{\mathrm{ads}}$ to it; this is a rule for the inputs, not a claim that $\Gamma_{\mathrm{Ca,sn}} = 0$.

0.6 Nucleation on the template and the sign convention

Physics. A cluster gains $\Delta\mu$ per formula unit but pays interfacial free energy for every unit of area it creates; the barrier is the top of that trade-off. On a substrate the cluster replaces part of its surface with a cheaper substrate contact, which lowers the barrier without changing the driving force.

A sphere of radius r with molecular volume $v_{\mathrm{m}} = V_{\mathrm{m}}/N_{\mathrm{A}}$ has $\Delta G_{\mathrm{hom}}(r) = -\frac{4\pi r^3}{3}\Delta\mu/v_{\mathrm{m}} + 4\pi r^2\gamma_{\mathrm{nl}}$, so

$$r^* = \frac{2\gamma_{\mathrm{nl}} v_{\mathrm{m}}}{\Delta\mu} \quad [\mathrm{m}], \qquad \Delta G^*_{\mathrm{hom}} = \frac{16\pi\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\Delta\mu^2} \quad [\mathrm{J}]. \tag{S14}$$

A spherical cap of contact angle θ (measured through the nucleus) on a flat substrate has $\Delta G_{\mathrm{het}}(r) = -V_{\mathrm{cap}}\Delta\mu/v_{\mathrm{m}} + \gamma_{\mathrm{nl}}A_{\mathrm{cap}} + (\gamma_{\mathrm{sn}} - \gamma_{\mathrm{sl}})A_{\mathrm{base}}$. Stationarity in θ at fixed volume is Young's relation, and substituting it factorises the whole free energy:

$$\begin{aligned} \cos\theta &= \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}}{\gamma_{\mathrm{nl}}}, \qquad \gamma_{\mathrm{sn}} = \gamma_{\mathrm{sn}}^{(0)} + \Delta\gamma_{\mathrm{w}}; \\ \Delta G^*_{\mathrm{het}} &= f(\theta)\Delta G^*_{\mathrm{hom}}, \qquad f(\theta) = \frac{(2+\cos\theta)(1-\cos\theta)^2}{4} \quad [1;\ \mathrm{J}]. \end{aligned} \tag{S15}$$

$f(0) = 0$, $f(\pi/2) = \frac{1}{2}$, $f(\pi) = 1$ and $f' = \frac{3}{4}\sin^3\theta \geq 0$, so $0 \leq \Delta G^*_{\mathrm{het}} \leq \Delta G^*_{\mathrm{hom}}$. The factorisation needs one and the same γ_{nl} on the cap surface and in Young, so the bare γ_{nl} stays in the denominator.

The water-displacement term. Displacing the template's bound water under the contact disc costs free energy over the contact only: $\Delta\gamma_{\mathrm{w}} \equiv \gamma_{\mathrm{sn}} - \gamma_{\mathrm{sn}}^{(0)} > 0$ is a cost, placed on the substrate side. Then $\partial f/\partial\Delta\gamma_{\mathrm{w}} = 3\sin^2\theta/(4\gamma_{\mathrm{nl}}) \geq 0$: a cost raises the barrier. Writing $\gamma_{\mathrm{eff}} = \gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}}$ in the denominator instead would give $\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = (\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}^{(0)})/\gamma_{\mathrm{eff}}^2 > 0$ on the wetting side, so a cost would lower the barrier, and it would break the factorisation. If γ_{sn} is a full in-water energy, $\Delta\gamma_{\mathrm{w}} \equiv 0$.

Consequence of binding for the barrier. With $\gamma_{\mathrm{sl}} = \gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}}$,

$$\begin{aligned} \cos\theta &= \frac{\gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}} - \gamma_{\mathrm{sn}}^{(0)} - \Delta\gamma_{\mathrm{w}}}{\gamma_{\mathrm{nl}}}, \\ \frac{\partial\cos\theta}{\partial\Delta G_{\mathrm{seq}}} &= \frac{\Gamma_{\mathrm{s}}\Theta}{\gamma_{\mathrm{nl}}} > 0 \quad [1;\ \mathrm{mol\,J^{-1}}]. \end{aligned} \tag{S16}$$

At fixed $\gamma_{\mathrm{sl}}^{(0)}$ and γ_{sn}, stronger binding lowers $\cos\theta$ and raises the barrier: ions held at the template–liquid interface must be stripped from the area the nucleus covers, the physics of an adsorbed inhibitor. Binding promotes nucleation only if γ_{sn} falls by more than γ_{sl}, that is, by Eq. (S13), only if the contact holds more Ca^{2+} excess than the template–liquid interface. Lowering γ_{nl} lowers f only when $\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}$.

0.7 No double counting

Physics. The template is both an interface and a ligand field, so its binding could be counted twice: once as a lower interfacial energy and once as a higher driving force. The open system at fixed bulk chemical potentials has one free energy, and writing it down shows where each effect sits.

Forming a cap of radius r transfers $n = V_{\mathrm{cap}}/v_{\mathrm{m}}$ formula units from the reservoir at the bulk chemical potentials, creates the cap surface and replaces template–liquid by template–nucleus interface over the base. The grand potential change is

$$\Delta\Omega_{\mathrm{G}}(r) = -n\,k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}} + \gamma_{\mathrm{nl}}A_{\mathrm{cap}} + (\gamma_{\mathrm{sn}} - \gamma_{\mathrm{sl}})A_{\mathrm{base}} \quad [\mathrm{J}], \tag{S17}$$

with grand-canonical interfacial energies at the bulk chemical potentials. Within capillarity this is exact; the reservoir term carries no template dependence, and by Gibbs' construction every effect of the template on the ions near it is an excess per area. Its stationary value is the screen composite

$$\begin{aligned}\Delta G^{*}_{\mathrm{het}} = f\Big(\theta\big[\gamma^{(0)}_{\mathrm{sl}} + \Delta\gamma_{\mathrm{ads}}(\Delta G_{\mathrm{seq}}, c_{\mathrm{bulk}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}}),\ \gamma^{(0)}_{\mathrm{sn}},\ \Delta\gamma_{\mathrm{w}}\big]\Big) \\ \times \frac{16\pi\gamma^{3}_{\mathrm{nl}}v^{2}_{\mathrm{m}}}{3(k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}})^{2}} \quad [\mathrm{J}].\end{aligned} \tag{S18}$$

Statement. In Eq. (S18) every effect of the template on the barrier is counted exactly once.

Proof. (1) *Exactness.* Eq. (S18) is the stationary value of the exact grand potential (S17) after the factorisation of Section 0.6, with $\gamma_{\mathrm{sl}} = \gamma^{(0)}_{\mathrm{sl}} + \Delta\gamma_{\mathrm{ads}}$. (2) *Channel separation.* Ω_{bulk} depends on bulk activities only, so $\partial\Omega/\partial(\text{any template input}) = 0$; all template dependence is in γ_{sl} and γ_{sn}, hence in θ. γ_{nl}, V_{m} and T are template-independent and may appear in both factors. (3) *Single entry of the binding.* ΔG_{seq}, Γ_{s} and δ_{s} enter γ_{sl} through Eq. (S12) only, and only when γ_{sl} is built from $\gamma^{(0)}_{\mathrm{sl}}$. They do not enter γ_{sn}, supplied from contact data with its own excess inside, and they do not enter Ω (Eq. (S4)). $\Delta\gamma_{\mathrm{w}}$ enters γ_{sn} once. (4) Hence each template-dependent input appears once, in one interfacial energy, and only through θ. This completes the proof.

The two errors. The local-ideal bookkeeping gives the enriched ion the bulk standard state, $\Omega_{\mathrm{li}} = (c_{\mathrm{local}}/c_{\mathrm{bulk}})\Omega_{\mathrm{bulk}}$, so that the layer is treated as a reservoir held above the bulk; with Ca-free interfacial energies it counts the binding once, in a channel that does not exist at equilibrium, and with the opposite effect on the barrier. Combined with the Ca-loaded γ_{sl} it counts the binding twice. Both exceed the exact driving force by $k_{\mathrm{B}}T\ln(c_{\mathrm{local}}/c_{\mathrm{bulk}})$ per formula unit, $-\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$ in the dilute limit, and both lower the barrier by the factor $[\ln\Omega_{\mathrm{bulk}}/\ln\Omega_{\mathrm{li}}]^{2}$. A second application of $\Delta\gamma_{\mathrm{w}}$ is the third double count. The implementation keeps these composites only as named demonstrations.

0.8 Admissibility of the template–liquid energy

Physics. An interfacial free energy is the work of making unit area of interface. A template film whose interface with water had a negative energy would lower its free energy by making more interface. It would swell or disperse until chain elasticity, crosslinks or anchoring stopped it, and the bounded interface that the cap free energy assumes would not exist. The adsorption term has no such limit built in, so the model must impose one.

The requirement. At every registered state,

$$\gamma_{\mathrm{sl}} = \gamma^{(0)}_{\mathrm{sl}} + \Delta\gamma_{\mathrm{ads}} \geq 0 \quad [\mathrm{J\,m^{-2}}]. \tag{S19}$$

It is necessary, not sufficient, and an ultra-low positive γ_{sl} is admitted. With $\gamma_{\mathrm{sn}} \geq 0$ and $\gamma_{\mathrm{nl}} > 0$ it makes all three in-water energies non-negative. Its Ca-free case is $\gamma^{(0)}_{\mathrm{sl}} \geq 0$. $\gamma^{(0)}_{\mathrm{sl}}$ must be the same interface without Ca^{2+} at the regime temperature, pH and ionic strength. An energy measured with Ca^{2+} at c_{ref} needs only $\Delta\gamma_{\mathrm{ads}}(c_{\mathrm{bulk}}) - \Delta\gamma_{\mathrm{ads}}(c_{\mathrm{ref}})$ added; adding $\Delta\gamma_{\mathrm{ads}}(c_{\mathrm{bulk}})$ in full double-counts. An energy measured in pure water or estimated from probe liquids on the dry film lacks the pH and

ionic-strength terms. By the Lippmann relation those lower the energy of a film charged further from its point of zero charge, so the sourced value overstates the reference. The adsorption term is a small-change expansion about the Ca-free interface. Its validity ratio $r_{\mathrm{ads}} = -\Delta\gamma_{\mathrm{ads}}/\gamma_{\mathrm{sl}}^{(0)}$ is reported, and Eq. (S19) is $r_{\mathrm{ads}} \leq 1$.

A solution constant is not a site free energy. With the site-volume standard state, a complexation constant K_{CaL} referenced to c° gives $\Delta G_{\mathrm{site}} = -RT\ln[K_{\mathrm{CaL}}\Gamma_{\mathrm{s}}/(\delta_{\mathrm{s}}c^\circ)]$ and the Langmuir odds $x = \alpha_2 K_{\mathrm{CaL}} c_{\mathrm{bulk}}/c^\circ$. For a constant that refers to the catecholate dianion, the cycle must charge both protons:

$$\Delta G_{\mathrm{seq}} = \Delta G_{\mathrm{site}} + RT\ln\bigl(1 + 10^{\,\mathrm{p}K_{\mathrm{a}2}-\mathrm{pH}} + 10^{\,\mathrm{p}K_{\mathrm{a}}+\mathrm{p}K_{\mathrm{a}2}-2\mathrm{pH}}\bigr) \quad [\mathrm{J\,mol^{-1}}]; \tag{S20}$$

the one-proton cycle overstates the binding. Lateral repulsion within a layer of fixed sites and competition by protons and cations lower the excess at every concentration. Because $\Pi_{\mathrm{ads}} = -\Delta\gamma_{\mathrm{ads}} = RT\int \Gamma_{\mathrm{Ca}}\,\mathrm{d}\ln c$, the Langmuir term at a given affinity is an upper bound on the lowering with respect to them. Site geometry moves the affinity either way by an unknown amount, so a verdict built on the constant stays conditional on it.

The bounds on Γ_{s} and δ_{s} do not cap the term below $\gamma_{\mathrm{sl}}^{(0)}$. Since $\partial\Pi_{\mathrm{ads}}/\partial c_{\mathrm{bulk}} = (RT/c_{\mathrm{bulk}})\Gamma_{\mathrm{Ca}}$, the lowering is largest where the excess vanishes, $c_0 = c_{\max}(1 - 1/\mathcal{E})$:

$$\Pi_{\mathrm{ads}} \leq RT\Gamma_{\mathrm{s}}\Big(\ln\mathcal{E} - 1 + \frac{1}{\mathcal{E}}\Big) < -\Gamma_{\mathrm{s}}\Delta G_{\mathrm{seq}} \quad [\mathrm{J\,m^{-2}}], \tag{S21}$$

independent of δ_{s} and of $\gamma_{\mathrm{sl}}^{(0)}$. With $\Gamma_{\mathrm{s}} = 1/(N_{\mathrm{A}}a_{\mathrm{site}})$, some dose overshoots once $\ln\mathcal{E} - 1 + 1/\mathcal{E} > \gamma_{\mathrm{sl}}^{(0)}a_{\mathrm{site}}/k_{\mathrm{B}}T$. For a hydrophilic film the right side is a fraction of one, so binding by a few $k_{\mathrm{B}}T$ suffices.

The rule and its consequences. The joint prior of $(\gamma_{\mathrm{sl}}^{(0)}, \Delta G_{\mathrm{seq}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}})$ is restricted to Eq. (S19) at every state and design point. Refused draws are counted, and the retained mass is reported beside every verdict that uses γ_{sl}. A saturating adsorption term is rejected: the lowering is bounded by the site layer, not by $\gamma_{\mathrm{sl}}^{(0)}$, and a floor at zero would invent a boundary state. Clipping $\cos\theta$ at -1 is rejected too, because it counts a non-existent interface as non-wetting. The admissible set is an upper set in ΔG_{seq} ($\partial\gamma_{\mathrm{sl}}/\partial\Delta G_{\mathrm{seq}} = \Gamma_{\mathrm{s}}\Theta > 0$), so the refused draws are the strongest binders. The single entry of the binding and its sign are unchanged. Finally, $f \leq \frac{1}{2}$ exactly when $\gamma_{\mathrm{sn}} \leq \gamma_{\mathrm{sl}}$, and a binding template has $\gamma_{\mathrm{sl}} \leq \gamma_{\mathrm{sl}}^{(0)} + RT\delta_{\mathrm{s}}c_{\mathrm{bulk}}$. A favourable template therefore needs a crystal–template contact at least as favourable as the template–water contact. This holds under any prior, and binding narrows the region.

0.9 Wetting branches, the spreading coefficient and the ignorance prior on γ_{sn}

Physics. Whether the mineral spreads on the template or beads into islands is a wetting balance. When no measurement of the template–nucleus contact exists, its admissible range is set by stability of a sharp interface; the range says what thermodynamics allows, not where the true value lies.

Spreading. Replacing template–liquid interface by a thick layer of mineral changes the free energy per area by

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} = \gamma_{\mathrm{nl}}(\cos\theta - 1) \quad [\mathrm{J\,m^{-2}}], \tag{S22}$$

the second form on the branch where Young has a solution. $S_{\mathrm{spread}} \geq 0$ is complete wetting (layer growth, $f = 0$); $S_{\mathrm{spread}} < 0$ is island growth.

Bounds on γ_{sn}. Separating the template–nucleus contact into two surfaces against vacuum costs $W_{adh,dry} = \gamma_p + \gamma_n - \gamma_{sn}$, which cannot be negative: otherwise the solids would separate spontaneously and no contact would exist. A negative interfacial free energy between immiscible solids would make the system create contact without limit, so $\gamma_{sn} \geq 0$. Hence

$$0 \leq \gamma_{sn} \leq G \equiv \gamma_p + \gamma_n \quad [\mathrm{J\,m^{-2}}], \tag{S23}$$

with γ_n the dry surface energy of the framing phase. The wet work $\gamma_{sl} + \gamma_{nl} - \gamma_{sn}$ is not bounded below: it is negative exactly on the non-wetting branch, and water can displace a nucleus. The same two arguments give $\gamma_{ow} \in [0, \gamma_o + \gamma_w]$ and $\gamma_{op} \in [0, \gamma_o + \gamma_p]$ (Section 0.1). The ignorance prior is uniform on the admissible range.

Branch masses under the prior. Young splits the γ_{sn} axis at $a = \gamma_{sl} - \gamma_{nl}$ (complete wetting below) and $b = \gamma_{sl} + \gamma_{nl}$ (non-wetting above). For γ_{sn} uniform on $[0, G]$ and the other energies fixed,

$$\begin{gathered} P_{\mathrm{complete}} = \mathrm{clip}\Big(\frac{\gamma_{sl} - \gamma_{nl}}{G}\Big), \qquad P_{\mathrm{non}} = \mathrm{clip}\Big(1 - \frac{\gamma_{sl} + \gamma_{nl}}{G}\Big), \\ P_{\mathrm{partial}} = 1 - P_{\mathrm{complete}} - P_{\mathrm{non}} \quad [1], \end{gathered} \tag{S24}$$

with clip to $[0, 1]$; when both cuts lie in $[0, G]$, $P_{\mathrm{partial}} = 2\gamma_{nl}/G$. Writing $\gamma_{sl} = \gamma_p + \gamma_w - W_{pw}$ and $\gamma_{nl} = \gamma_n + \gamma_w - W_{nw}$, the non-wetting mass is positive iff $W_{pw} + W_{nw} > 2\gamma_w$: together the two solids bind water more strongly than water binds itself. The marginal masses are expectations over the priors of γ_{sl} and γ_{nl}. They are properties of the prior and are never reported as a probability of feasibility; flip values and required precisions are reported instead.

Complete wetting is excluded on a hydrophilic template. Because $\gamma_{sn} \geq 0$,

$$S_{\mathrm{spread}} \leq \gamma_{sl} - \gamma_{nl} \quad [\mathrm{J\,m^{-2}}], \tag{S25}$$

so complete wetting needs $\gamma_{sl} \geq \gamma_{nl}$. Here a template is called hydrophilic when water wets it at least as favourably, per area, as water wets the mineral: $\gamma_{sl} \leq \gamma_{nl}$. This is not the usual criterion of a small water contact angle in air, which by Young in air, $\cos\theta_{air} = (\gamma_p - \gamma_{sl})/\gamma_w$, constrains $\gamma_p - \gamma_{sl}$ and does not place γ_{sl} below γ_{nl}. For a template that is hydrophilic in this sense, $S_{\mathrm{spread}} \leq -\gamma_{sn} < 0$ for every admissible $\gamma_{sn} > 0$: complete wetting is excluded whatever the template–nucleus contact is, the cut point a lies at or below zero and $P_{\mathrm{complete}} = 0$. This is a physical result, not a property of the prior. Ca^{2+} binding lowers γ_{sl} further ($\Delta\gamma_{ads} < 0$), so the mechanism's own premise pushes the same way. Whether a given template meets the condition is a question for the inputs, so $P(\gamma_{sl} > \gamma_{nl})$, the largest mass the spreading route can carry, is reported at every state.

Coalescence. With $S_{\mathrm{spread}} < 0$ a film forms only where islands meet. Equal caps of angle θ on a square lattice of density N_s cover the cells at the cap height $h_{cl} = \tan(\theta/2)/\sqrt{2N_s}$ (first contact at $h_c = h_{cl}/\sqrt{2}$), so closure below an admissible thickness $d_{\max}$ requires

$$\begin{gathered} N_s \geq N_{s,cl} = \frac{\tan^2(\theta/2)}{2d_{\max}^2} \quad [\mathrm{m^{-2}}], \\ d_{\max} = \max\Big(\frac{R_{A,\max}}{\rho_e},\ \frac{\ln(j_0/j_{req})}{2\kappa}\Big), \quad \kappa = \frac{\sqrt{2m_e\phi_B}}{\hbar} \quad [\mathrm{m}], \end{gathered} \tag{S26}$$

$d_{\max}$ is the larger of the ohmic and direct-tunnelling bounds for an insulating film that must pass the collector current; it is a limitation, not a performance figure. The lattice is the optimistic arrangement, and the island density is taken as the upper estimate N_{nuc} of the next part. A continuous film is admitted through either route: complete wetting, or closure below $d_{\max}$.

0.10 The absolute nucleation criterion (C3)

Physics. A barrier is low enough only relative to the attempts the system makes in the time available. What must be achieved is set by geometry: a cathode particle touches the collector through a contact patch, and a patch with no nucleus under it is a patch the mineral cannot cover from below.

Nuclei form on the template at the rate per area $J = J_0 \mathrm{e}^{-\Delta G^*_{\mathrm{het}}/k_{\mathrm{B}}T}$, with J_0 the kinetic prefactor per area. Along the registered R-slurry states k, held for Δt_k, the expected areal density is

$$N_{\mathrm{nuc}} = \sum_k J_0 \Delta t_k \, \mathrm{e}^{-\Delta G^*_{\mathrm{het},k}/k_{\mathrm{B}}T} \quad [\mathrm{m}^{-2}], \tag{S27}$$

an upper estimate, since it ignores exclusion zones around growing islands; a state with $\Omega_{\mathrm{bulk}} \le 1$ contributes nothing. If nuclei land independently, the number under one patch of area A_{patch} is Poisson distributed with mean $N_{\mathrm{nuc}} A_{\mathrm{patch}}$, and the fraction of patches left empty is $\mathrm{e}^{-N_{\mathrm{nuc}} A_{\mathrm{patch}}}$. Requiring that at most a fraction q_0 be empty,

$$\mathrm{e}^{-N_{\mathrm{nuc}} A_{\mathrm{patch}}} \le q_0 \iff N_{\mathrm{nuc}} \ge N_{\mathrm{req}} = \frac{\ln(1/q_0)}{A_{\mathrm{patch}}} \quad [\mathrm{m}^{-2}], \qquad q_0 = 1 - p, \tag{S28}$$

with p the credible level of the decision rule (p = `[NumPcred]`, swept from `[NumPcredSweepLow]` to `[NumPcredSweepHigh]`), so the admissible empty fraction moves with it. At a single state the criterion is

$$\Delta G^*_{\mathrm{het}} \le B_{\mathrm{acc}} = k_{\mathrm{B}}T \ln \frac{J_0 t_{\mathrm{p}}}{N_{\mathrm{req}}} = k_{\mathrm{B}}T \ln \frac{J_0 t_{\mathrm{p}} A_{\mathrm{patch}}}{\ln(1/q_0)} \quad [\mathrm{J}]. \tag{S29}$$

Only $\ln J_0$ enters, so an order-of-magnitude prefactor moves B_{acc} by a few $k_{\mathrm{B}}T$, and the expectation form, one nucleus per patch on average, is the special case $q_0 = 1/\mathrm{e}$. N_{req} is fixed apart from the closure density of C4, so that C3 does not borrow the island geometry of C4.

The patch area. The part of a particle touching a flat foil lies inside its projection, and the particle touches through at least one asperity contact, so

$$a_{\mathrm{asp}} \le A_{\mathrm{patch}} \le \frac{\pi d_{\mathrm{p}}^2}{4}, \qquad \frac{4\ln(1/q_0)}{\pi d_{\mathrm{p}}^2} \le N_{\mathrm{req}} \le \frac{\ln(1/q_0)}{a_{\mathrm{asp}}} \quad [\mathrm{m}^2;\ \mathrm{m}^{-2}], \tag{S30}$$

with d_{p} the projected-area-equivalent particle diameter. The smaller patch is the strict end. The bound spans many decades and only its logarithm enters, so the prior is loguniform. While either end is unsourced the criterion is reported as a gap, and the flip values of A_{patch} (where the good mass equals p and $1-p$) are reported in its place. The wetting part of C3 asks the complementary question, whether the template prefers the nucleus to the liquid: $f \le \frac{1}{2}$, $\theta \le \pi/2$, the only value in $(0, 1)$ at which the sign of the Young numerator changes.

0.11 Thermodynamic stability (C5)

Physics. A layer that is unstable in its environment dissolves or reacts, and the two regimes are different environments. In the slurry the question is whether $CaCO_3$ survives alkaline water; in the cell, whether the phases survive the collector's potential against the species the electrolyte offers. Both are screens: necessary, not sufficient, and blind to kinetics and morphology.

R-slurry: dissolution is decided by Ω_{bulk}. Carbon is held in the C(IV) carbonate system (the regime fixes the carbonate activity, and reduction of dissolved carbonate is not observed in aqueous processing; the construction with carbon redox allowed is a labelled sensitivity). Dissolving the layer is then $CaCO_3(s) \rightarrow Ca^{2+} + CO_3^{2-}$ at the regime activities, with free energy $RT \ln K_{\mathrm{sp}} - RT \ln(a_{Ca^{2+}} a_{CO_3^{2-}}) = -RT \ln \Omega_{\mathrm{bulk}}$ per mole. The layer dissolves when $\Omega_{\mathrm{bulk}} < 1$ and is stable

against dissolution when $\Omega_{\mathrm{bulk}} \geq 1$, with the measured calcite K_{sp} and the speciated activities of Section 0.3. This is the same Ω_{bulk} that drives nucleation: a state that cannot nucleate $CaCO_3$ is a state in which the layer dissolves, and by Section 0.2 enrichment at the template changes neither. As a Pourbaix hull distance per non-O, non-H atom ($n_{\mathrm{M}} = 2$), with the dissolved ions on the hull,

$$\Delta G_{\mathrm{pbx}}\big|_{\mathrm{diss}} = \frac{RT}{n_{\mathrm{M}}} \ln \frac{1}{\Omega_{\mathrm{bulk}}} \quad (\Omega_{\mathrm{bulk}} < 1) \quad [\mathrm{J\,mol^{-1}}]. \tag{S31}$$

A computed hull therefore cannot decide dissolution. Its tolerance $\tau = n_{\mathrm{at}} e_{\mathrm{f}} / n_{\mathrm{M}}$ reports dissolution only below $\ln \Omega_{\mathrm{bulk}} = -n_{\mathrm{at}} e_{\mathrm{f}} / (RT)$, which hides several decades of undersaturation, and an error δ of the computed dissolution free energy shifts the apparent K_{sp} by $\delta / (RT \ln 10)$ decades. The test is applied at every design point of total calcium and dissolved carbon (the dose window) and must report dissolution at an undersaturated control state and stability at a supersaturated one; otherwise it is inconclusive.

R-slurry: competing solids. The Pourbaix construction

$$\begin{aligned} \Psi_\phi = \frac{1}{n_{\mathrm{M}}} \Big[& G_\phi - n_{\mathrm{O}} \mu_{\mathrm{H_2O}} - RT \ln 10\, (2n_{\mathrm{O}} - n_{\mathrm{H}}) \mathrm{pH} \\ & - (2n_{\mathrm{O}} - n_{\mathrm{H}} + Q) F E_{\mathrm{SHE}} \Big], \\ \Delta G_{\mathrm{pbx}} = \Psi_\phi - \Psi_{\mathrm{hull}} \geq 0 \quad & [\mathrm{J\,mol^{-1}}], \end{aligned} \tag{S32}$$

with C(IV) fixed and the dissolved Ca species removed, answers the remaining question: whether another solid lies below $CaCO_3$, within the tolerance of computed formation energies. Adding the amorphous offset to the edge makes the amorphous-framing test equal to the calcite-level test, because the offset shifts Ψ and leaves the hull unchanged. The native oxide's state is a baseline, not a gate.

R-cell: the window from explicit oxidation reactions. Holding the collector at E_{Li} fixes $\mu_{\mathrm{Li}} = \mu^\circ_{\mathrm{Li}} - F E_{\mathrm{Li}}$. A reaction that releases Δn Li to the reservoir has $\Delta\Phi = \Delta G_{\mathrm{r}} - \Delta n F E_{\mathrm{Li}}$ and sets the limit

$$\begin{aligned} E^* &= \frac{\Delta G_{\mathrm{r}}}{\Delta n F} \quad [\mathrm{V}]; \\ E_{\mathrm{ox}} = \min_{\Delta n > 0} E^*, &\qquad E_{\mathrm{red}} = \max_{\Delta n < 0} E^*. \end{aligned} \tag{S33}$$

A Li-free phase has no Li-releasing reaction of its own, so a construction open in Li alone declares $CaCO_3$ stable at any anodic potential by construction. Anodic stability needs reactions that hand the carbonate's electrons to the reference and send the cation into what the electrolyte offers: Li^+, the salt anion A^- and the solvent. Electroneutrality pairs each released cation with A^-, and the electrons plate Li from LiA:

$$\begin{aligned} \mathrm{CaCO_3} + 2\,\mathrm{LiA} &\rightarrow \mathrm{CaA_2} + \mathrm{CO_2} + \tfrac{1}{2}\mathrm{O_2} + 2\,\mathrm{Li}, \\ \mathrm{Al_2O_3} + 6\,\mathrm{LiA} &\rightarrow 2\,\mathrm{AlA_3} + \tfrac{3}{2}\mathrm{O_2} + 6\,\mathrm{Li}, \\ 2\,\mathrm{AlOOH} + 6\,\mathrm{LiA} &\rightarrow 2\,\mathrm{AlA_3} + \mathrm{H_2O} + \tfrac{3}{2}\mathrm{O_2} \\ &\quad +6\,\mathrm{Li}, \end{aligned} \tag{S34}$$

for the three gated phases ($CaCO_3$, corundum and boehmite), each with $\Delta n > 0$.

The sink logic. The sink is fixed by the registered electrolyte, not chosen: only its anions are admissible, and the oxidation limit is the minimum over admissible sinks and products, since a phase is unstable if any accessible reaction is downhill. The true product is the lower in free energy of the solid salt and the solvated ions, so a limit computed with the solid salt is an upper bound on the true onset. A falsification computed with it stands; a pass computed with it is inconclusive,

and a pass counts only with the solvated sink. A sink whose thermochemistry is missing makes the limit undetermined, which never passes. The electrolyte's own anodic limit is a separate quantity that the screen does not assess.

Which thermochemistry decides. A computed reaction energy that combines database solids with a gas reference carries the error of both, and the reference schemes bracket the measured energies from opposite sides. Where every reactant and product is tabulated, the all-experimental route has no such error and governs, with margin m_{expt}: the tabulated uncertainties per transferred electron plus a bound on any neglected solid entropy (for the registered reaction with the perchlorate sink, $m_{\mathrm{expt}} =$ `[NumMexpt]`). A computed scheme corrected by its measured benchmark error reproduces that same route, so it checks the bookkeeping and is never a second confirmation. Where the experimental route is missing, the computed schemes decide with margin $m_E = n_{\mathrm{unc}} e_{\mathrm{f}}/(\Delta n F)$, counting the atoms whose computed energy does not cancel ($m_E =$ `[NumMwindow]` for the registered reaction): a pass must hold under the lowest-onset scheme and a falsification under every scheme. A margin belongs to one reaction and one sink. The governing rule is

$$\begin{gathered} \text{falsified if } E_{\mathrm{ox}}^{\mathrm{expt}} + m_{\mathrm{expt}} < E_{\mathrm{cell}}^{\mathrm{max}} \text{ (either sink);} \\ \text{feasible only if } E_{\mathrm{ox}}^{\mathrm{expt}} - m_{\mathrm{expt}} \geq E_{\mathrm{cell}}^{\mathrm{max}} \\ \text{(solvated sink),} \end{gathered} \tag{S35}$$

mirrored for reduction against $E_{\mathrm{cell}}^{\mathrm{min}}$. The charge cutoff $E_{\mathrm{cell}}^{\mathrm{max}}$ belongs to the cathode, so the verdict is a function of it, with crossovers at $E_{\mathrm{ox}}^{\mathrm{expt}} \pm m_{\mathrm{expt}}$. The reading of the limit as the onset of an observable reaction is supported by the analogous carbonate: $\mathrm{Li_2CO_3} \rightarrow 2\mathrm{Li}{+}\mathrm{CO_2}{+}\frac{1}{2}\mathrm{O_2}$ is measured to oxidise near the experimental equilibrium limit of the same reaction, so a solid carbonate in a non-aqueous cell oxidises close to its thermodynamic limit, with little kinetic overpotential.